FIBRED IN GROUPOIDS IS TRANSITIVE ALONG COMPOSABLE FUNCTORS

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Editorial note. Counted span is the author's own proof environment, source lines 7679-7691, following the statement at 7671-7677.

Editorial note. The proof is complete; every step is justified by a cited lemma.

Allowed standard prerequisites (editorial).

- Criterion for fibred in groupoids. lemma-fibred-groupoids, cited at proof lines 7681 and 7689-7690. Supplies (a) that every morphism of A is strongly B-cartesian and (b) the groupoid criterion restated in terms of fibre groupoids.
- Fibred over fibred is fibred. lemma-fibred-over-fibred, cited at proof line 7682. Supplies that A is fibred over C, reducing the claim to the groupoid property of the fibres A_U.
- Strongly cartesian morphisms with invertible image are invertible. lemma-composition-cartesian, cited at proof lines 7688-7689. Supplies that a strongly B-cartesian morphism whose image in B is an isomorphism is itself an isomorphism.

Source reference. Original lemma 02XL: sources/stacks-categories/pinned-original-source.tex, source lines 6459–6459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 003V: sources/stacks-categories/pinned-original-source.tex, source lines 7253–7253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09WV: sources/stacks-categories/pinned-original-source.tex, source lines 6852–6852. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A} \to \mathcal{B} \to \mathcal{C}$ be functors between categories. If $\mathcal{A}$ is fibred in groupoids over $\mathcal{B}$ and $\mathcal{B}$ is fibred in groupoids over $\mathcal{C}$, then $\mathcal{A}$ is fibred in groupoids over $\mathcal{C}$.*

Proof. One can prove this directly from the definition. However, we will argue using the criterion of Lemma 003V. By Lemma 09WV we see that $\mathcal{A}$ is fibred over

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$\mathcal{C}$. To finish the proof we show that the fibre category $\mathcal{A}_U$ is a groupoid for U in $\mathcal{C}$. Namely, if $x \to y$ is a morphism of $\mathcal{A}_U$, then its image in $\mathcal{B}$ is an isomorphism as $\mathcal{B}_U$ is a groupoid. But then $x \to y$ is an isomorphism, for example by Lemma 02XL and the fact that every morphism of $\mathcal{A}$ is strongly $\mathcal{B}$-cartesian (see Lemma 003V). $\square$